Financial Market Effects of FOMC Communication: Evidence from a New Event-Study Database*

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Abstract

This paper introduces the U.S. Monetary Policy Event-Study Database (USMPD), a novel, public, and regularly updated dataset of high-frequency financial market data around Federal Open Market Committee (FOMC) policy announcements, press conferences, and minutes releases. Using the rich data in the USMPD, we document several new empirical findings. Large monetary policy surprises have made a comeback in recent years, and post-meeting press conferences have become the most important source of policy news. Monetary policy surprises have pronounced negative effects on breakeven inflation based on Treasury yields. Risk assets, including dividend derivatives, also respond strongly and negatively to monetary policy surprises, consistent with conventional channels of monetary transmission. Press conferences have stronger effects than FOMC statements on most asset prices, and we recommend the use of monetary event surprises that include both. Finally, the term structure evidence shows peak effects on market-based inflation and dividend expectations at horizons of several years.

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1 Introduction

High-frequency market movements around central bank announcements, or "monetary policy surprises", are widely used to estimate the causal impact of monetary policy on financial markets, expectations, and the macroeconomy. However, a unified reference source of monetary policy surprises for the Federal Reserve does not exist.

This paper establishes the U.S. Monetary Policy Event-Study Database (USMPD), a public, transparent, and regularly updated dataset of high-frequency asset price changes around official FOMC communication events.[1] The database collects intraday market reactions measured around four different events: FOMC statements, post-meeting press conferences, FOMC meeting minutes, and a "monetary event" that includes both the statement and the press conference. The USMPD covers a broad range of assets, including various money market futures rates, OIS rates, Treasury and TIPS yields, U.S. stock market indices and dollar exchange rates. It contains all series required to produce the most commonly used measures of monetary policy surprises, and to estimate the intraday market response of interest rates and risk assets to news about the Fed's monetary policy.

Leveraging the expanded data coverage of communication events, financial instruments, and sample periods in the USMPD, we establish new evidence on the transmission of U.S. monetary policy to financial markets. Inspired by Nakamura and Steinsson (2018), we calculate monetary policy surprises for each of the four different types of events as the normalized first principal component of changes in money market futures rates covering a horizon of about one year. These measures capture both target rate news from the policy decisions announced in FOMC statements and news about the future rate path communicated via statements, press conferences, and minutes.

Our first set of results pertains to the features of monetary policy surprises, both over time and across events. The magnitude of monetary policy surprises has increased markedly since the beginning of the tightening cycle in March 2022. In other words, monetary policy surprises

[1] The USMPD is available at https://sffed.us/USMPD.

have made a comeback, and using them in empirical analysis is both more promising and more relevant than the pre-pandemic decline in surprise volatility suggested. Post-meeting press conferences emerge as a primary source of policy news, largely independent of the statements and having recently surpassed them in information content. The correlation between the two types of surprises has varied notably over time: modestly positive before the 2022 liftoff and negative since, as press conferences have tended to counteract the shift in rate expectations caused by the statement release. By contrast, while still informative about the policy path, FOMC meeting minutes generate much less market volatility: Most relevant policy information appears to be communicated effectively via the FOMC statements and the Chair's press conferences. As a practical matter, we therefore recommend the monetary event surprise, which combines information about both the statements and associated press conferences, as a simple yet comprehensive summary measure of FOMC policy news.

We then use these high-frequency monetary policy surprises to study the effects of FOMC communication on financial markets. For Treasury markets, our main new finding is that monetary policy surprises have significant negative effects on the term structure of inflation compensation. The inclusion of a more recent sample relative to the earlier literature (Hanson and Stein, 2015; Nakamura and Steinsson, 2018) is key to this result, which reconciles the financial market evidence with the time series evidence of disinflationary effects of monetary policy shocks (Gertler and Karadi, 2015; Miranda-Agrippino and Ricco, 2021; Bauer and Swanson, 2023a). Across events and using the extended sample, we confirm that the response of stock prices, exchange rates, and other risk-sensitive indicators is generally strong and directionally consistent with the transmission of monetary policy shocks: the prices of risk assets tend to respond negatively to monetary policy surprises. For both Treasuries and risk assets, we find that the market response is especially strong to policy news from post-meeting press conferences. Both the magnitude of the effects and the explanatory power tend to be larger than for the commonly used statement surprises.

Using data on dividend futures and swaps, we present new evidence on the effects of

FOMC surprises on (risk-adjusted) dividend expectations. Prices for dividend claims tend to respond negatively to policy surprises, especially at longer horizons. Our estimates are directionally consistent with standard New Keynesian models that imply negative effects of monetary policy on future output and profits (Bilbiie and Känzig, 2023), and with the typically sluggish response of dividends to news (Lintner, 1956; Leary and Michaely, 2011). By contrast, earlier studies by Golez and Matthies (2025) and Gilchrist et al. (2024) have estimated positive effects on short-run dividend strips, due to differences in horizons and measurement.

For both inflation and dividends, our term structure evidence shows the strongest effects on market-based expectations at longer horizons of several years. If our policy surprises mainly captured short-rate policy shocks, these patterns would imply significant expected lags and very persistent effects for the monetary transmission to inflation and real activity. But high-frequency monetary policy surprises capture a broader concept of policy news, and an alternative explanation seems more plausible: The effects on market-based expectations can be long-lasting because the signals from monetary policy surprises affect investor beliefs about the future conduct of monetary policy, including the perceived inflation target, the natural rate of interest, and responsiveness to economic conditions. This interpretation is consistent with recent work on the effects of monetary policy on the perceived reaction function (e.g., Bauer et al., 2024b; Bocola et al., 2025).

Our contributions have broad relevance for macro-finance research. The USMPD provides a new reference database of high-frequency market changes around FOMC communication events that other researchers can readily use. Our empirical evidence uncovers new results for the effects of monetary policy on Treasury markets and risk assets, and contributes to open debates on key issues in monetary economics: the sensitivity of Treasury yields, inflation and dividend expectations to monetary policy surprises, the empirical importance of different types of central bank communication, the risk-taking channel of monetary policy, information effects, and lags in the monetary transmission. We hope that both the database and our empirical results will stimulate further research on these policy-relevant topics.

Our paper is related to several strands of the monetary economics literature. From a methodological standpoint, our paper follows the successful model of event-study databases published for other central banks, including the European Central Bank (Altavilla et al., 2019), the Bank of England (Braun et al., 2025), and the Swedish Riksbank (Almerud et al., 2024). Our approach for constructing high-frequency surprises around FOMC communication events builds on the seminal papers by Kuttner (2001) and Gürkaynak et al. (2005), and we incorporate insights and best practices from recent work on monetary policy surprises, including Acosta et al. (2024) and Brennan et al. (2025).

Since the pathbreaking work of Kuttner (2001), Bernanke and Kuttner (2005), and Gürkaynak et al. (2005), a large and growing literature has used high-frequency identification to estimate the effects of the Fed's monetary policy on financial markets both in the U.S. and abroad.[2] We revisit and resolve several open issues in this literature. Some early studies of how firmly long-run inflation expectations are anchored estimated negative, though not always statistically significant, responses of long-horizon inflation compensation to Kuttner surprises (Beechey and Wright, 2009; Beechey et al., 2011; Gürkaynak et al., 2010). More recently, Hanson and Stein (2015) and Nakamura and Steinsson (2018) investigated the response of the Treasury market to FOMC statement surprises and found essentially no response of breakeven inflation rates. We document that either extending the sample period or incorporating FOMC press conferences yields the negative effects on market-based inflation expectations one would expect based on standard macro models and time series estimates of monetary transmission. Golez and Matthies (2025) and Gilchrist et al. (2024) estimated positive short-run effects of Fed policy surprises on dividend strip prices, while we show that evidence from longer-maturity dividend swaps and futures is consistent with standard channels

[2]See Gürkaynak et al. (2010), Kroencke et al. (2021), Swanson (2021), Miranda-Agrippino and Nenova (2022), Bauer et al. (2023), and Cieslak and McMahon (2023), among many others. In addition, see Campbell et al. (2012); Nakamura and Steinsson (2018); Bauer and Swanson (2023b) for high-frequency estimates of the effects of monetary policy on survey-based expectations, and Faust et al. (2004); Gertler and Karadi (2015); Jarociński and Karadi (2020); Miranda-Agrippino and Ricco (2021); Bauer and Swanson (2023a) for estimates of the macroeconomic effects.

of monetary transmission. Recent work by Nagel and Xu (2026) suggests that changes in risk-free discount rates play an important role for the negative stock market response to monetary policy surprises. Our results show that effects on the dividend expectations and risk premia, as captured by dividend swap prices, also contribute materially to this response.

Only a few papers have studied market reactions to other FOMC communication events beyond FOMC statements. Most closely related to our work is the study by Swanson and Jayawickrema (2024), which investigates the financial market effects of FOMC announcements, press conferences, minutes releases, and policy-relevant speeches by the Fed Chair and Vice Chair. Swanson and Jayawickrema show that Chair speeches lead to higher market volatility than FOMC announcements, and document the significant response of Treasury yields and the S&P 500 to policy surprises constructed for these events. Our work complements theirs, with new results on the empirical patterns of policy surprises and the effects on the term structures of market-based inflation and dividend expectations.[3]

The paper is structured as follows. Section 2 introduces the USMPD and its content. In Section 3 we explain how we construct monetary policy surprises from these data and present new facts about the policy news from different FOMC communication events. Section 4 shows results for Treasury markets with a focus on breakeven inflation rates. Section 5 turns to risk assets, including stocks, exchange rates, corporate bond spreads, and dividend derivatives. Section 6 concludes. An Online Appendix provides details about the USMPD and the construction of policy surprises, and contains additional empirical results.

2 The USMPD: A New Database for FOMC Events

The USMPD collects high-frequency changes around FOMC communication events for a range of interest rates and asset prices.[4] It is available on the website of the Federal Reserve Bank of San Francisco and updated after every FOMC meeting. This section describes the

[3]Other papers studying FOMC communication beyond the post-meeting statement release include Swanson (2023), Narain and Sangani (2026), Bauer and Swanson (2023a), and Cieslak et al. (2024).

[4]We obtain the underlying intraday tick data from LSEG Tick History, formerly Refinitiv Tick History.

content of the database, including the FOMC communication events, the construction of high-frequency rate changes and asset returns over different intraday event windows, and the different instruments contained in the database. Details are in Online Appendix A.

2.1 FOMC Communication Events

The FOMC communicates to the public through several channels. After each of the eight annual scheduled policy meetings, the *FOMC statement* describes the Committee's current economic assessment, policy decisions, and future outlook for the economy and monetary policy. Statements were first released in 1994, and the USMPD contains data for every meeting since then, including the meetings from 1994 to 1999 that were not followed by the release of a statement. Market data for unscheduled FOMC meetings—usually conference calls that take place during times of financial stress—is included as well. At roughly every other policy meeting, the statement is accompanied by the Summary of Economic Projections (SEP), which summarizes the economic forecasts of FOMC meeting participants. The USMPD identifies those meetings with an SEP release.

Post-meeting *press conferences* by the Fed Chair were introduced in 2011, and held after every meeting since 2019. During a press conference, the Chair's opening remarks are followed by a Q&A session during which the Chair addresses questions from reporters. The Chair speaks on behalf of the FOMC during these press conferences. The USMPD includes the market reaction to all press conferences.

The Fed regularly publishes the *minutes* of FOMC meetings. Since 2005, the meeting minutes have been released to the public three weeks after each meeting. Before that, the minutes were released with a longer delay and after the subsequent FOMC meeting, which makes those observations less useful for event studies. The USMPD includes market data for the minutes releases starting in 2001.

Speeches by Fed governors and Bank presidents are another important channel of communication about monetary policy. However, they differ in several important ways from FOMC communication events. First, speeches reflect the views of individual policymakers and not those of the Committee, as the usual disclaimer makes clear. Second, many speeches focus on other topics beyond monetary policy, including financial regulation, payment systems, or community engagement.[5] Third, many speeches are less suitable for financial market event studies, as they can be long, delivered after trading hours, or without timestamps. Because of these differences, speeches are not included in the USMPD. We refer interested readers to the work by Swanson and Jayawickrema (2024).

For the empirical analysis in the paper, we use a sample that ends in December 2024. This sample includes 265 FOMC meetings with 81 press conferences. Our sample of minutes releases contains 160 observations over the period from 2005 to 2024.

2.2 Intraday Event Windows

Monetary policy event studies rely on the assumption that changes in asset prices during tight windows around policy announcements are predominantly driven by the revealed policy news. If the event windows are too wide, many other factors may affect asset prices, making statistical inference more difficult. By contrast, if the windows are too narrow, market participants may not have enough time to fully digest the news, implying that prices may not fully incorporate the new information.

For FOMC statements, we follow common practice since Gürkaynak et al. (2005) and use 30-minute windows. Our "statement window" therefore starts 10 minutes before and ends 20 minutes after the release of the statement. For FOMC meeting minutes, we use the same approach as for FOMC statements. While the minutes are much longer than the FOMC statements, they are also released to reporters under embargo, and the available evidence suggests rapid price discovery (Rosa, 2013). The "minutes window" is the 30-minute interval that starts 10 minutes before and ends 20 minutes after the release of the minutes. The post-meeting press conferences by the Fed Chair usually last between 45 minutes and one

[5] Swanson and Jayawickrema (2024) found that fewer than half of the speeches given by the Chair and Vice Chair over the period from 1988 to 2023 contained information about monetary policy.

hour. Thus we define the "press conference window" as a 70-minute interval that starts 10 minutes before and ends 60 minutes after the beginning of the press conference.

The USMPD also includes market data for a "monetary event window" that combines information in the FOMC announcement and the post-meeting press conference. For FOMC meetings without a press conference, this window is identical to the 30-minute statement window. For those meetings followed by a press conference, the monetary event window instead ends 60 minutes after the beginning of the press conference. The market changes over the monetary event window capture all of the monetary policy communication on days with FOMC meetings.

2.3 Interest Rates and Asset Prices in the USMPD

For each of the four different event windows, the USMPD contains high-frequency changes for various interest rates, stock market indices, and exchange rates, mainly based on intraday data from LSEG Tick History. More specifically, the USMPD includes high-frequency *changes* for various interest rates: federal funds, Eurodollar and Secured Overnight Financing Rate (SOFR) futures; overnight index swap (OIS) rates; and yields on Treasury bonds and Treasury Inflation-Protected Securities (TIPS). These are calculated as the simple difference in the pre- and post-event (implied) interest rates, in percentage points. For stock market indices and dollar exchange rates, the USMPD includes high-frequency *returns*, calculated as the relative price change, in percent, over each FOMC communication event window.

Money market futures, and in particular federal funds futures, have long been used to gauge financial markets' expectations about current and future policy rates, going back to the seminal work by Kuttner (2001). The USMPD includes high-frequency changes for the first six monthly contracts, FF1 to FF6. Each of these provides an estimate of changes in market expectations for the average funds rate over the contract month. It is straightforward to calculate the implied surprise change in the policy rate for the current FOMC meeting, denoted by MP1, and the change in the expected rate for the next meeting, denoted by

MP2 (Gürkaynak et al., 2005). These two implied rate surprises are commonly used in the construction of monetary policy news for FOMC announcements, and we extend this logic to other FOMC communication events.

For market expectations about interest rates at longer maturities, the empirical literature typically used Eurodollar futures, quarterly contracts settled based on the 3-month London Interbank Offered Rate (LIBOR) prevailing at the expiration date. These highly liquid money market futures provided an estimate of market-based short rate expectations up to a horizon of several years. The USMPD includes data for eight quarterly Eurodollar futures contracts which expire at the end of each of the next eight quarters relative to the FOMC event date, denoted by ED1 to ED8. We deal with the phasing out of LIBOR and Eurodollar futures by following the recommendation of Acosta et al. (2024) and switching from Eurodollar futures to SOFR futures starting in January 2022. To retain consistency with the futures data up to 2021, the USMPD contains data for eight SOFR futures that align with the first eight quarterly Eurodollar futures.[6]

The USMPD further contains high-frequency changes in various Treasury interest rates: three- and six-month T-Bill rates, (nominal) Treasury yields with maturities of 2, 5, 10 and 30 years, and (real) TIPS yields with maturities of 5, 10 and 30 years. These are all based on intraday observations for the benchmark yields from on-the-run securities. The database also includes one- and two-year OIS rates.

Besides interest rates, the database also includes high-frequency returns on several risk assets. For the stock market, we include returns in the S&P 500 index, the S&P 500 E-mini futures and the Wilshire 5000 index. Finally, we include the dollar index DXY and bilateral exchange rates of the U.S. dollar against the euro and the Japanese yen.

[6]For measuring monetary policy expectations, SOFR futures have the important advantage that their prices depend on overnight interest rates in the repo market, which are closely related to the federal funds rate, whereas LIBOR rates and thus Eurodollar futures were also affected by changes in credit risk. A SOFR futures contract references the quarter *ending* at its expiration, whereas a Eurodollar contract references the quarter *beginning* at its expiration, so ED1 corresponds to the second, rather than the first, active quarterly SOFR futures contract, and so on. See Online Appendix A.4.3 for details.

3 Monetary Policy Surprises Across FOMC Events

In this section, we describe the construction of monetary policy surprises across the different FOMC communication events included in the USMPD. We document the properties of these policy surprises, including their changing behavior in recent years. We offer some guidance for the use of these measures in empirical research.

3.1 Construction of FOMC Surprises

The empirical approach of constructing policy surprises as changes in money market futures rates around FOMC announcements goes back to Kuttner (2001), Bernanke and Kuttner (2005), and Gürkaynak et al. (2005). These changes can be viewed as "surprises" under the assumption that over such short windows, interest rate changes are essentially unpredictable.[7] Various choices and combinations of near-term interest rates have been used in the measurement of monetary policy surprises. The information included in the USMPD allows researchers to construct any measure of their choice, including the policy surprises proposed by Gürkaynak et al. (2005), Gertler and Karadi (2015), Jarociński and Karadi (2020), Swanson (2021), and Bauer and Swanson (2023a). Online Appendix B describes the construction of some key alternative measures from the data in the USMPD.

In our analysis, we follow the approach of Gürkaynak et al. (2005) and Nakamura and Steinsson (2018) for constructing FOMC announcement surprises and extend it to other communication events. These studies used 30-minute changes around FOMC statement releases in MP1, MP2, ED2, ED3, and ED4, which together cover a horizon of roughly one year. To parsimoniously capture the policy news about both the current and expected future policy rate for each of our four communication events we use these same five rate changes,

[7]Several recent studies have documented some predictability using economic news and financial data observed before the FOMC announcements, at least based on in-sample regressions (Miranda-Agrippino, 2016; Cieslak, 2018; Bauer and Swanson, 2023b). While this "ex post" predictability can have important consequences for high-frequency identification in monetary VARs, it does not seem to cause issues for financial market event studies for FOMC announcements (Bauer and Swanson, 2023a). In this paper, we follow common terminology and denote as "surprise" any high-frequency rate changes around FOMC events, and leave the issue of predictability to future research.

which we summarize using a single principal component as in Nakamura and Steinsson (2018). By contrast, Gürkaynak et al. (2005) derived two separate "target" and "path" factors.

Our procedure delivers four different measures of monetary policy surprises: for FOMC statements, press conferences, the combined monetary event, and the minutes. A principal component, however, is only defined up to a scaling factor, and to obtain an economically interpretable quantity, we normalize each surprise as in Nakamura and Steinsson (2018) so that it has a one percentage point effect on the daily change in the one-year Treasury yield on average over the sample. Online Appendix B reports the loadings of the five underlying components on each of these surprise measures, and explains how other types of surprises can be calculated from the data in the USMPD.

The surprises for FOMC statements, which correspond directly to those used by Nakamura and Steinsson (2018), capture news about the average expected policy rate over the subsequent year. The statement surprise is effectively an average of the target and path factors of Gürkaynak et al. (2005), corresponding to a *level factor* of rate changes. Empirically, this type of surprise tends to be highly correlated with interest rates at longer maturities—as evident from Nakamura and Steinsson (2018) and our results in Section 4—and should therefore not be interpreted as conventional, short-run monetary policy news but as reflecting the impact of monetary policy statements on the average policy path, which can include shifts at long horizons. Indeed, the calculation is agnostic about the type of announcement and treats all types of news equally, such as news about changes in the federal funds rate, forward guidance, or QE.[8]

Press conferences and minutes releases contain only information about the future path of the policy rate, for example via forward guidance. The surprises around these events are therefore akin to a *slope factor* of near-term rate changes. The surprise around the monetary event window combines news from both the statements and the press conferences,

[8]Calculating measures that separately identify target rate, forward guidance, or QE surprises can be a powerful tool in different empirical applications (Gürkaynak et al., 2005; Swanson, 2021, 2024; Jarociński, 2024). Note that news about QE can affect near-term expectations for the policy rate path via signaling effects (Bauer and Rudebusch, 2014).

and therefore tends to include relatively more information about the rate outlook than the statement surprise alone. For the sample of policy decisions in this paper, there are only 81 press conferences on the 265 days with FOMC announcements. Going forward, with press conferences occurring after every FOMC meeting, the information revealed over the monetary event window will naturally be tilted even more towards forward-looking information.

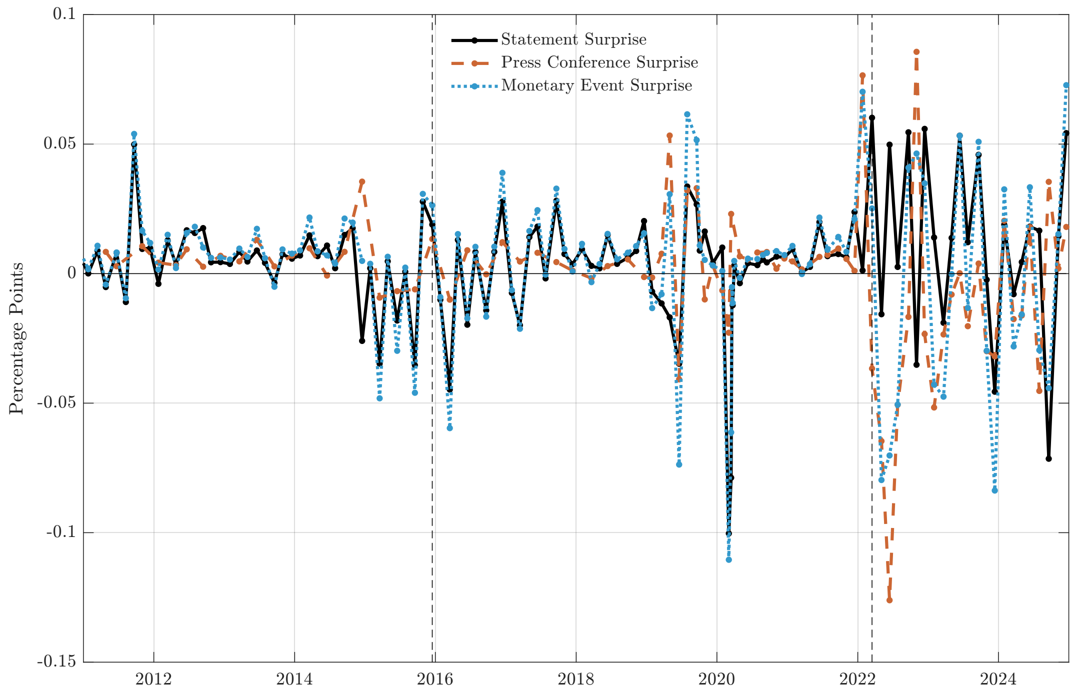
3.2 Monetary Policy Surprises Since 2011: Four New Facts

The most recent evidence and expanded set of communication events in the USMPD reveal several new facts about the behavior of monetary policy surprises. Our focus is on the period since 2011, when post-meeting press conferences were first introduced. Figure 1 plots the time series of surprises around FOMC statements, press conferences, and monetary events, over this period. Table 1 reports volatility of monetary policy surprises across subsamples. Online Appendix B provides further details.

First, large monetary policy surprises have made a comeback. As Table 1 shows, the period since liftoff in March 2022 has been marked by sizable surprises, with statement volatility rising to 3.5 bp and press conference volatility exceeding 4 bp. During the earlier period shown in Figure 1, magnitudes were noticeably smaller not only during zero-lower-bound episodes but also from 2015 to 2019, when statement volatility was under 2 bp despite the policy rate being unconstrained. Previous studies, including Ramey (2016) and Swanson (2021), have noted this pattern of declining surprise volatility, which began well before the financial crisis and has been attributed to increased Fed transparency and forward guidance. Over the full sample from 1994 to 2024, statement volatility was 3.7 bp, driven largely by the pre-ZLB period when it reached 4.8 bp. The return of large policy surprises strengthens the case for high-frequency identification in empirical monetary economics.

Several forces likely contributed to this resurgence: elevated macroeconomic uncertainty and changes in the Fed's monetary policy framework, which prompted learning about its updated reaction function (Bauer et al., 2024a; Cieslak et al., 2024), uncertainty about

Figure 1: Time series of monetary policy surprises across FOMC events



Monetary policy surprises (in percentage points) around FOMC statements, press conferences, and monetary events from 2011 to 2024. Vertical lines indicate liftoff dates December 16, 2015 and March 16, 2022.

inflation persistence (Hajdini et al., 2025), and unusually large macroeconomic shocks, met by a policy response that departed sharply from standard policy rules (Nakamura et al., 2025). While some of these forces were specific to the post-pandemic episode, the broader lesson is that sudden macroeconomic shifts can generate large policy surprises even when the central bank is transparent.[9]

Second, the introduction of press conferences since 2011 has added a powerful additional source of monetary policy news. The volatility of these surprises has been similar to, or at times exceeded, that of statement surprises. Meetings with press conferences also saw larger statement surprises than those without: among the 81 meetings with a press conference since April 2011, statement volatility was 2.8 bp compared to only 1.1 bp for the 35 meetings

[9]Online Appendix B shows that the level of interest rates does not appear to be an important driver of the volatility of monetary policy surprises. Outside of ZLB periods, a regression of the absolute statement surprise on the lagged one-year yield has an insignificant coefficient and an R^2 of essentially zero, and the largest individual surprises have occurred across very different rate environments.

Table 1: Volatility of monetary policy surprises across subsamples

			Standard Deviation				Correl.
	Start	End	STMT	PC	ME	MIN	(STMT,PC)
Full sample	1994	2024	3.73	–	4.09	1.43	–
Since PC	Apr-2011	2024	2.41	2.73	3.05	1.10	−0.14
Pre-liftoff	Apr-2011	Jan-2022	2.02	1.60	2.50	0.92	0.23
Post-liftoff	Mar-2022	2024	3.52	4.07	4.67	1.62	−0.27
Pre-ZLB	1994	2008	4.76	–	5.03	2.35	–
Between ZLB	2015	2019	1.88	1.77	2.72	1.12	0.44

Summary statistics of monetary policy surprises—around FOMC statements (STMT), press conferences (PC), monetary events (ME), and minutes (MIN)—for different sample periods: volatilities (in basis points) and pairwise correlation between statement and press conference surprises. Press conference surprises are available starting April 2011; minutes surprises starting 2005.

without one. The presence of a press conference thus amplifies the scope for policy surprises around FOMC statements, likely because those meetings involved more consequential policy decisions. The importance of news from press conferences has increased even further in recent years. While before the liftoff in 2022 press conference volatility of 1.6 bp was below statement volatility of 2.0 bp, since then press conferences have caused larger policy surprises than the FOMC statements (4.1 vs. 3.5 bp).

Third, press conferences constitute an additional source of information about monetary policy that is largely independent of the information conveyed in policy statements. Over the whole sample from 2011 to 2024, statement and press conference surprises have been essentially uncorrelated. This is what one would expect, since the FOMC statement is released before the press conference begins and markets incorporate statement news quickly.

Table 1 shows, however, that this correlation varies substantially over time. Before the 2022 liftoff, the correlation was modestly positive, indicating that press conference surprises tended to reinforce the direction of statement news. The period between the two ZLB episodes, from 2015 to 2019, saw the highest correlation of 0.44. Since liftoff in 2022, by contrast, statement and press conference surprises have been negatively correlated: several large hawkish statement surprises in 2022 and 2023 were followed by dovish press conference surprises on the same day.[10] Over the full sample since April 2011, these shifts roughly cancel out, leaving the unconditional correlation close to zero. In sum, disregarding press conferences when analyzing the effects of monetary policy risks missing a sizable and largely independent portion of relevant policy news.

Fourth, since 2011 the release of the minutes has generated less than half the market volatility of the statements and about a third that of the monetary event, making the minutes the least informative type of FOMC communication event. Effective communication would imply exactly this: the FOMC discloses most of the relevant information about monetary policy promptly, through its standard channels after the meeting. In any event, the small footprint of the minutes releases should be kept in mind when interpreting subsequent regression results.[11]

These facts about the role of press conferences and the limited importance of minutes releases suggest that measures of monetary policy surprises should, at a minimum, account for the information revealed in both the FOMC statements and the post-meeting press conferences.[12] A simple and effective way to do this is by using the *monetary event window* that covers both events. Figure 1 shows that the resulting monetary event surprise tracks the statement news quite closely in the early period, and since 2022 is instead more heavily influenced by the press conferences. Overall, we view the monetary event surprise as an accurate and parsimonious summary of monetary policy news on FOMC meeting days, and recommend it to researchers looking for a simple measure of Fed policy surprises.

[10] Narain and Sangani (2026) document a similar reversal in stock and Treasury market responses, and use textual analysis to link it to the Chair's word choice in press conferences.

[11] The normalization that we adopt to translate the different types of news into interpretable units means that the minutes news is scaled by a much larger factor to yield a 1 ppt daily change in the one-year nominal yield.

[12] Swanson and Jayawickrema (2024) also document the growing importance of press conferences and the limited market impact of minutes releases, based on the volatilities of money market futures rates.

4 Treasury Yields and Breakeven Inflation

This section documents new evidence on the sensitivity of nominal Treasury yields, TIPS yields, and breakeven inflation (BEI) rates to monetary policy surprises. Earlier research, going back to Kuttner (2001) and Gürkaynak et al. (2005), has studied the effects of monetary policy on the Treasury market in detail, with important insights. We add several novel results to this literature, by extending the event-study analysis to other communication events beyond FOMC statements and updating it to include more recent monetary policy cycles and the post-COVID inflationary episode.

For most of the following analysis, we use daily data for nominal Treasury yields from Gürkaynak et al. (2007) and for TIPS yields from Gürkaynak et al. (2010), all measured in percent. BEI rates, also called inflation compensation rates, are differences of nominal and real (TIPS) yields, thus capturing market-based expectations of inflation over the maturity of the yields.[13] While data on Treasury yields extend back for many decades, the TIPS market did not come into existence until 1997, and exhibited low liquidity until the early 2000s. To account for this, and for comparability across yields, we restrict our analysis in this section to estimation samples starting in 2004, when the TIPS market had left its infancy and provided reliable information about real interest rates. Our sample period ends in 2024. Given the collapse in TIPS market liquidity during the financial crisis, and following Nakamura and Steinsson (2018), we exclude the period from July 2008 to June 2009 throughout this section. For all our event studies here and below in Section 5, we also exclude six unscheduled FOMC meetings (one with a press conference) that happened during or following a market closure, so that the daily event window spans a weekend or holiday.[14] The final sample includes 167 FOMC announcements (both scheduled and unscheduled), 80 press conferences, and 152

[13]Daily yield data is available at https://www.federalreserve.gov/data/yield-curve-models.htm. These smoothed zero-coupon curves flexibly capture the entire term structure of interest rates without being affected by idiosyncratic, instrument-specific variation and noise, such as on-the-run premia or repo specialness spreads.

[14]For such events, the multi-day window likely contains the market developments that prompted the policy action, raising identification concerns. Online Appendix Table C.4 reports estimates under alternative treatments of unscheduled meetings.

minutes releases.

Our analysis relies on event-study regressions of the form

$$\Delta y_t = \alpha + \beta\, s_t + \varepsilon_t, \tag{1}$$

where t indexes days with FOMC communication events, and $\Delta y_t = y_t - y_{t-1}$ denotes the change in a generic interest rate, such as nominal, real, and BEI yields or forward rates of a certain maturity. To facilitate comparison with the earlier literature, our baseline results use daily rate changes calculated using end-of-day bond prices as dependent variables. The USMPD also includes high-frequency changes for several Treasury and TIPS yields, and we report additional results using intraday regressions in Online Appendix C.[15] For each communication event, the regressor s_t denotes the related monetary policy surprise as constructed in Section 3 and scaled to yield a 1 ppt increase in the 1-year nominal yield over the full available sample. The parameter β quantifies the effect of monetary policy news across FOMC events on different interest rates. As long as the information in s_t is predetermined with respect to the financial market data on day t, such event-study regressions estimate the causal effects of the Fed's monetary policy actions and communication on asset prices.[16]

4.1 Effects of FOMC Announcements on Treasury Markets

We begin our analysis with the study of monetary policy news around FOMC announcements. For this event window, our statement surprise is the same measure of Fed policy news as in Nakamura and Steinsson (2018).

[15] Online Appendix Table C.6 reports the regressions of Table 2 for two-day event windows: Our conclusions are unchanged, and the negative responses of BEI rates tend to be more pronounced than over the baseline one-day windows.

[16] This assumption is plausibly satisfied for statement and minutes surprises. For surprises including the press conference, it is possible that the Chair's communication could be affected by previous market changes in the day, in which case s_t would only be predetermined in regressions with intraday changes as the dependent variable. However, we show in Online Appendix C that regressions for intraday changes in Treasury yields lead to very similar results, suggesting that this effect does not lead to any noticeable bias in event-study regressions for daily market changes.

Table 2: Response of Treasury market to different monetary policy surprises

	5y yld				5-10y forw			
	STMT	PC	ME	MIN	STMT	PC	ME	MIN
(A) Nominal								
Coefficient	0.96	1.58	1.30	1.12	−0.19	0.47	0.03	0.38
	(0.28)	(0.25)	(0.22)	(0.31)	(0.20)	(0.16)	(0.16)	(0.33)
R^2	0.11	0.30	0.28	0.08	0.01	0.05	0.00	0.01
N	167	80	167	152	167	80	167	152
(B) Real								
Coefficient	1.17	1.99	1.59	1.26	0.18	0.59	0.34	0.60
	(0.31)	(0.32)	(0.24)	(0.28)	(0.18)	(0.20)	(0.14)	(0.32)
R^2	0.13	0.30	0.35	0.12	0.01	0.08	0.03	0.03
N	167	80	167	152	167	80	167	152
(C) Inflation								
Coefficient	−0.21	−0.41	−0.30	−0.13	−0.37	−0.12	−0.31	−0.25
	(0.11)	(0.15)	(0.09)	(0.16)	(0.11)	(0.17)	(0.10)	(0.21)
R^2	0.02	0.08	0.07	0.00	0.06	0.01	0.06	0.01
N	167	80	167	152	167	80	167	152

Event-study regressions of daily changes in Treasury rates around FOMC communication events on high-frequency monetary policy surprises over different windows. STMT: 30-minute changes around FOMC statements. PC: changes around the post-meeting press conference. ME: changes over the monetary event window, that is, statements as well as press conferences when available. MIN: 30-minute changes around release of FOMC meeting minutes. Response of nominal Treasury rates in panel (A), of real (TIPS) rates in panel (B), and of breakeven inflation rates in panel (C). Rates are 5-year yields in the first four columns, and 5-to-10-year forward rate in the last four columns. Sample: for STMT and ME surprises, 167 FOMC announcements from 2004 to 2024; for PC surprises, 80 press conferences from 2011 to 2024; for MIN surprises, 152 releases of the minutes from 2005 to 2024; all sample periods excluding the financial crisis period from July 2008 to June 2009 and unscheduled FOMC meetings whose daily event window spans a market closure. White standard errors in parentheses.

Table 2 reports estimates of the response of five-year yields and 5-to-10-year forward rates to policy news, with the response of nominal, real, and breakeven inflation rates in the top, middle, and bottom panel, respectively. The columns labeled "STMT" correspond to the response to the FOMC statement surprise. Detailed estimates for the term structures of nominal, real and BEI rates are shown in Online Appendix C.

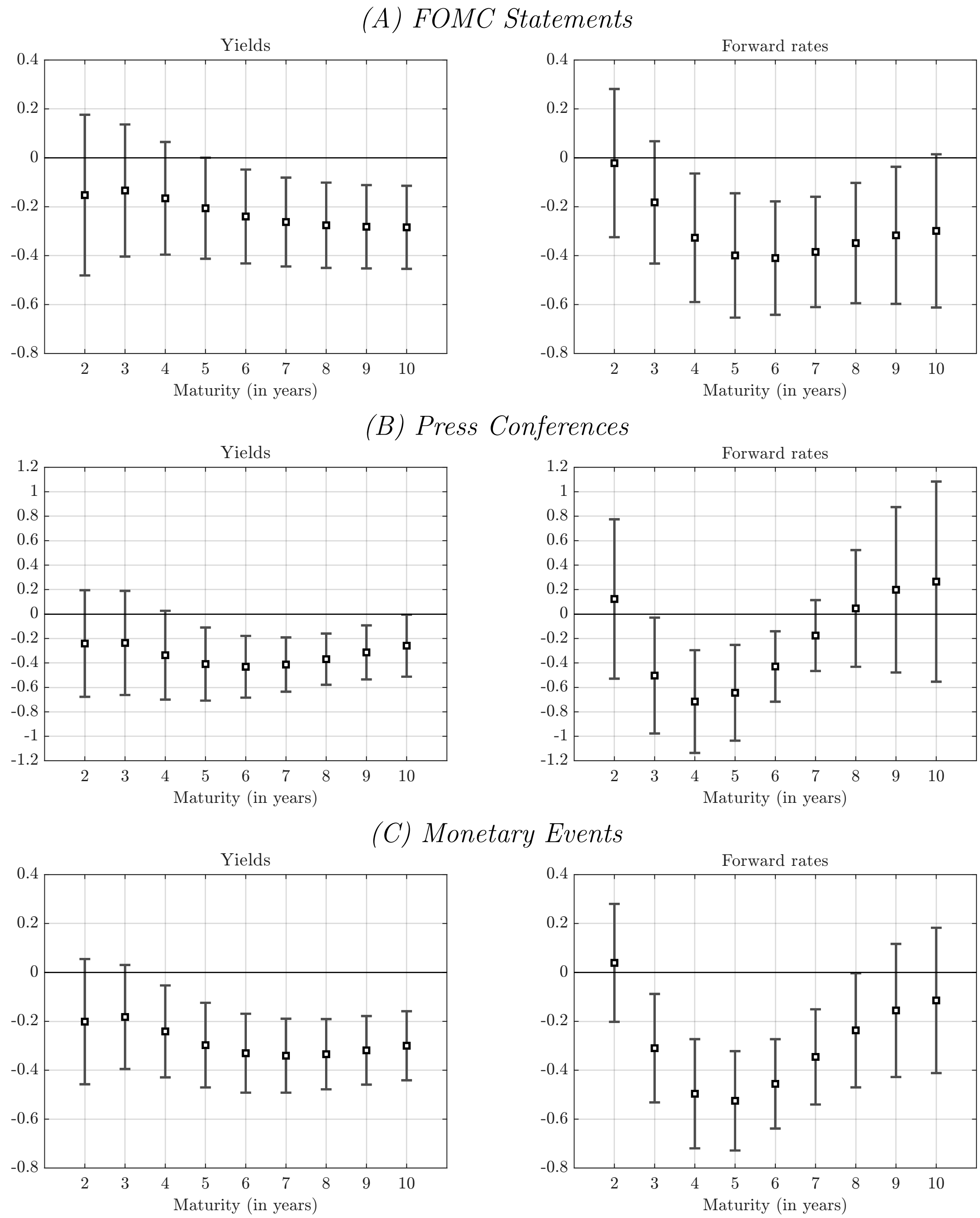
FOMC statement surprises significantly affect both nominal and real yields. This response and the pattern across maturities are well-documented. In our sample, the term structure response declines quickly enough with maturity such that long-term forward rates do not respond positively to FOMC statement surprises. That is, we do not find an "excess sensitivity puzzle" as in Hanson and Stein (2015) and Nakamura and Steinsson (2018).

The main new finding is the negative response of market-based inflation expectations. Statement surprises have a significantly negative impact on the five-year BEI yield. The responses of BEI rates across maturities to statement surprises are shown in Figure 2, for zero-coupon yields (top left panel) as well as for instantaneous forward rates (top right panel). BEI yields and forward rates at medium and longer maturities exhibit pronounced negative responses. The peak effect on forward rates is about -0.4, implying that a 10 basis point hawkish statement surprise on the one-year yield leads to a roughly 4 basis point decline in market expectations for inflation at these peak horizons.

Our findings contrast with the earlier evidence of Hanson and Stein (2015) and Nakamura and Steinsson (2018) that showed essentially no response of BEI rates at most horizons. That lack of policy sensitivity of market-based inflation expectations is puzzling from the perspective of standard New Keynesian models and monetary VARs, both of which indicate negative effects of monetary policy shocks on inflation (see, e.g., recent empirical work in Gertler and Karadi, 2015; Miranda-Agrippino and Ricco, 2021; Bauer and Swanson, 2023a). Our new results using an extended sample show the expected negative effects, which reconciles the estimated effects of policy surprises on inflation compensation with the direction in monetary macro models and time series evidence.

Breakeven inflation rates reflect market-based or "risk-adjusted" expectations, which include not only inflation expectations but also inflation risk premia and liquidity premia (D'Amico et al., 2018; Andreasen et al., 2021). Given the substantial challenges and uncertainty involved in the estimation of these premia, we focus our event-study analysis on observed market rates and do not analyze estimated expectations and risk premia. Evidence from New Keynesian macro-finance models (Hördahl and Tristani, 2012), as well as various empirical results in the macro-finance literature, suggests that inflation risk premia respond positively

Figure 2: Sensitivity of BEI rates to monetary policy surprises



Event-study regressions of daily changes in breakeven inflation (BEI) on high-frequency monetary policy surprises. Panel (A): responses to FOMC statement surprise, with 167 observations from 2004 to 2024. Panel (B): responses to press conference surprise, with 80 observations from 2011 to 2024. Panel (C): responses to monetary event surprise, including both FOMC statements and press conferences when available, with 167 observations from 2004 to 2024. Estimation excludes the financial crisis period from July 2008 to June 2009 and unscheduled FOMC meetings whose daily event window spans a market closure. Responses for BEI yields in left panels, for instantaneous BEI forward rates in right panels. Error bars show 95% confidence intervals based on White standard errors.

to monetary policy tightening.[17] In light of this evidence, the response of actual, real-world inflation expectations is likely even more negative than our estimated response of market-based inflation expectations.[18]

The estimated responses of forward BEI rates (top right panel of Figure 2) are hump-shaped, with a peak response at the five- to six-year horizon. The coefficients are significantly negative out to very long horizons, and Table 2 shows that the negative response of the 5-to-10-year forward BEI rate is highly significant. If monetary policy surprises mainly captured standard monetary policy shocks, then this term structure evidence could reflect long expected lags and highly persistent effects in the monetary transmission to inflation. But this interpretation appears at odds with the established time series evidence for the effects of monetary policy. Although estimates differ widely in their implications for transmission lags, this evidence generally suggests that peak effects of monetary policy on inflation occur at relatively short horizons of one to two years, and that the effects die out for longer horizons.[19]

The term structure patterns therefore suggest mechanisms that extend beyond a conventional monetary policy shock alone. If policy announcements contain signals about the conduct of monetary policy, this can explain the persistently negative responses of BEI rates. The idea that policy surprises can affect the perceived inflation target goes back to Gürkaynak et al. (2005), and recent studies have considered the effects on the coefficients in the perceived policy reaction function (Bauer et al., 2024a,b; Jarociński and Karadi, 2025). As an intuitive example, consider a hawkish statement surprise during a period of above-target

[17]Kim and Orphanides (2012) and Bauer (2018) document that estimates of risk premia in Treasury yields are highly uncertain and model-dependent. Evidence on the risk-taking channel of monetary policy indicates that policy surprises raise risk premia in financial markets (Bauer et al., 2023). Since most estimates of inflation risk premia tend to be positive (Christensen et al., 2010; Abrahams et al., 2016), an increase in their magnitude due to a hawkish policy surprise would increase the risk premium component of BEI rates. Evidence from Hanson and Stein (2015) and Gertler and Karadi (2015) on bond risk premia also points in this direction. Finally, monetary policy tightening tends to raise uncertainty (Mumtaz and Theodoridis, 2020; Bauer et al., 2022), a key determinant of risk premia (Wright, 2011).

[18]While TIPS liquidity premia could potentially confound our results, Online Appendix Figure C.4 shows that inflation swaps, which are unaffected by such premia, respond similarly to BEI rates.

[19]Examples include Christiano et al. (2005); Gertler and Karadi (2015); Jarociński and Karadi (2020); Miranda-Agrippino and Ricco (2021); Bauer and Swanson (2023a). Studies that point to more persistent effects and longer expected lags include Brunnermeier et al. (2021); Jordà et al. (2026); Aruoba and Drechsel (2024).

inflation. Such a policy surprise would not only raise near-term policy expectations, but also signal a lower long-run inflation target or a more aggressive inflation response, lowering longer-term inflation expectations due to revised beliefs about monetary policy. Bocola et al. (2025) show that with incomplete information about monetary policy, the sensitivity of long-term breakeven inflation to monetary policy surprises depends on the degree of anchoring of inflation expectations. Consistent with this prediction, Online Appendix C shows that BEI rate sensitivity was especially pronounced during the tightening episode 2015–2018, when inflation expectations may have been less anchored.

4.2 Press Conferences and Minutes

While empirical work on the financial market effects of Fed communication has generally focused on FOMC announcements, Section 3 showed that press conferences have become an important source of monetary policy news in recent years. We now provide new estimates of the effects of other communication events on Treasury markets.

Table 2 reports estimates for these three additional monetary policy surprises on the five-year yield and 5-to-10-year forward rate for nominal, real and BEI rates. Press conference surprises cause substantial responses of both real and nominal rates, and in most cases the estimated coefficients are materially larger in magnitude, and the R^2 substantially higher, compared to the estimates for the statement surprises. Press conference surprises have significantly positive effects on 5-to-10-year forward nominal and real rates, while statement surprises do not. In other words, we find excess sensitivity for press conference surprises, likely because press conferences contain policy news mainly in the form of forward guidance.

Press conference surprises have pronounced effects on medium-term BEI rates, and these effects are stronger than for statement surprises. The negative response of the five-year BEI yield is almost twice as large in magnitude, while the R^2 rises from 0.02 to 0.08. The middle panels of Figure 2 provide a detailed view of the response of the BEI term structure to press conference surprises. BEI yields respond negatively across all maturities, with a response that is statistically significant for maturities from 5 to 10 years. The negative response of BEI forward rates peaks around the 4-year horizon and then dies out and becomes statistically insignificant at the 7-year horizon. This term structure pattern is slightly different from that for statement surprises. Nevertheless, the significant negative responses at medium horizons still point to effects beyond the near term, consistent with the interpretation that surprise policy changes have longer-run effects by affecting beliefs about the conduct of monetary policy.

Our results suggest that press conference surprises have stronger effects on the Treasury market than the commonly studied statement surprises. Since our main results are based on univariate regressions over different sample periods, Online Appendix C carries this comparison further using several multivariate regressions which estimate the effects of statements and press conferences over the same sample and account for possible correlation between these surprises. We also report results for separate target and path factors derived from the statement surprises, following Gürkaynak et al. (2005), in order to directly compare the effects of forward guidance news from the statements and press conferences. These additional results show that for FOMC meetings with press conferences, the policy surprise around the press conferences has generally been the most important source of policy news for Treasury markets.

For the estimates using the monetary event surprises, Table 2 shows that in most cases, the coefficients are in between the estimates for the statement and press conference surprises. For the five-year nominal, real and BEI yields, the explanatory power of the monetary event surprise, measured in terms of R^2, is as strong as for the press conference surprise. For the 5-to-10-year forward BEI rates, the estimated effects of the monetary event surprises are almost as strong as those for the statement surprise. The bottom panels of Figure 2 show that monetary event surprises lead to peak responses in 4–6 year BEI forward rates. In general, the coefficients are estimated more precisely than for either statement or press conference surprises. The results suggest that the monetary event surprises effectively combine the information about the policy outlook and are therefore very useful for estimating the effects of monetary policy communication on financial markets.

Table 2 also shows the estimated response of the Treasury market to the release of the minutes of the FOMC meeting. This surprise, which captures news about the future policy rate path relative to the information previously communicated by the FOMC, leads to broadly similar responses of the five-year nominal and real yields as to the statement surprise. However, as noted in Section 3, minutes surprises are much less volatile, so the coefficient magnitude overstates the economic importance of a typical minutes release. Furthermore, there is no significant response of long-term real or nominal forward rates, or of any BEI rates, to the minutes surprise. Overall, the information in the FOMC meeting minutes has some moderate effects on medium-term Treasury yields but appears to be of limited quantitative importance for the Treasury market.

The event-study estimates in this section are based on daily zero-coupon Treasury yield curves. This evidence is somewhat conservative: Using the intraday yield changes contained in the USMPD produces even stronger results, as shown in Online Appendix C.

5 Risk Assets

Monetary policy is known to materially affect the prices of risk assets, including stocks, exchange rates, and corporate bond spreads (e.g., Bernanke and Kuttner, 2005; Jarociński and Karadi, 2020; Swanson, 2021; Bauer and Swanson, 2023a). These effects result from three main channels: First, changes in monetary policy affect the risk-free rates at which expected future cash flows are discounted, meaning that the effects documented in Section 4 are relevant for risk assets as well. Second, policy changes affect expectations of the risky cash flows underlying these assets, such as dividends and corporate defaults. Third, policy changes influence the risk premia that investors demand to bear the risks surrounding future cash flows, according to the risk-taking channel of monetary policy (Bauer et al., 2023). As an example, tighter policy would raise risk-free discount rates, lower dividend expectations and raise default expectations, and also lower the risk appetite of investors and financial intermediaries, thereby raising risk premia across a broad range of assets. All three of these channels would lead to lower prices of risk assets in response to tighter monetary policy, as captured by a hawkish monetary policy surprise. In addition to these "conventional" channels, monetary policy surprises could also lead to central bank information effects that would push in the opposite direction (Campbell et al., 2012; Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020). For example, a hawkish policy surprise may in part reflect a more benign economic outlook by the central bank, which could cause investors to revise their growth expectations upward. Therefore, in general, we would expect such information effects to attenuate the magnitude of the responses, and to potentially flip their sign if they dominate relative to conventional channels.[20] In the following analysis, we study the response of a broad set of risk asset prices to FOMC communication, using all communication events in our sample.

5.1 Stocks, Dollar, and Other Risk Assets

The first part of this analysis considers a range of assets and indicators that cover important markets, including stocks, exchange rates, corporate bonds, and mortgage rates. We include the following series: daily returns of the S&P 500 index, daily changes in the VIX, daily returns in the U.S. dollar index DXY, changes in an index of investor risk appetite, changes in the spread of the effective yield of the Bank of America BBB corporate bond index over the 10-year Treasury yield, changes in investment-grade CDS spreads, and changes in an MBS yield and an option-adjusted MBS spread.[21] We use one-day changes/returns for the

[20] Bauer and Swanson (2023a) provide evidence in favor of a "Fed response to news" channel that can also lead to a positive correlation between measured monetary policy surprises and changes in the economic outlook.

[21] The risk appetite measure is from Bauer et al. (2023) and ends in May 2022. For CDS spreads we use the CDX-IG index. Both MBS series are from Bloomberg: the yield is the Fannie Mae 30-year current-coupon yield (`MTGEFNCL Index`), and the spread is Bloomberg's option-adjusted spread for U.S. fixed-rate mortgage-backed securities (`LUMSOAS Index`), available only from August 2000.

Table 3: Response of risk assets to monetary policy surprises

	S&P 500	VIX	Dollar Index	Risk Appetite	BBB Spread	CDX IG	MBS Yield	MBS OAS
(A) FOMC Statements								
Coefficient	−10.0	10.0	4.1	−14.5	23.1	70.8	83.8	17.2
	(3.1)	(4.2)	(1.3)	(3.8)	(10.9)	(37.5)	(23.7)	(13.4)
R^2	0.08	0.03	0.08	0.08	0.01	0.10	0.06	0.01
N	259	259	259	238	235	169	259	206
(B) Press Conferences								
Coefficient	−15.6	20.5	7.7	−8.6	0.5	17.6	216.8	25.6
	(4.6)	(6.4)	(2.3)	(13.3)	(25.7)	(18.8)	(45.9)	(12.2)
R^2	0.12	0.05	0.16	0.01	0.00	0.01	0.21	0.02
N	80	80	80	59	80	80	80	80
(C) Monetary Events								
Coefficient	−10.9	11.5	4.7	−13.3	21.9	59.0	104.9	21.2
	(2.6)	(3.9)	(1.2)	(3.3)	(9.2)	(29.7)	(25.0)	(11.8)
R^2	0.12	0.05	0.13	0.08	0.01	0.10	0.11	0.02
N	259	259	259	238	235	169	259	206
(D) Minutes								
Coefficient	−3.3	−3.5	10.2	11.9	13.5	−68.0	95.7	−7.6
	(9.4)	(8.2)	(2.7)	(16.2)	(46.9)	(47.4)	(58.6)	(34.8)
R^2	0.00	0.00	0.10	0.01	0.00	0.03	0.02	0.00
N	160	160	160	140	160	160	160	160

Event-study regressions of changes/returns in risk assets around FOMC communication events on high-frequency monetary policy surprises over different windows. Panel (A): 30-minute changes around FOMC statements. Panel (B): changes around post-meeting press conferences. Panel (C): changes around statements and, when available, press conferences. Panel (D): 30-minute changes around release of FOMC meeting minutes. Risk assets: one-day return of S&P 500 (percent); one-day change in VIX; one-day return in dollar index DXY (percent); two-day change in risk appetite index; two-day change in BBB corporate bond spread (basis points); two-day change in investment-grade CDX index (basis points); two-day change in MBS yield and option-adjusted spread (OAS) (both basis points). Sample: FOMC meetings from February 1994 to December 2024 for panels (A) and (C); post-meeting press conferences from 2011 to 2024 for panel (B); all releases of the FOMC meeting minutes from 2005 to 2024 for panel (D). Unscheduled FOMC meetings whose daily event window spans a market closure are excluded. White standard errors in parentheses.

stock market, VIX, and dollar series, and two-day changes for all other series to account for slower price responses due to lower liquidity in corporate bonds and MBS markets. Returns are measured in percent, index changes in index points, and changes in yield and spreads in basis points.

Table 3 shows the event-study estimates. Around FOMC announcements, we find a strong and generally statistically significant response in the direction implied by conventional transmission channels. Stock prices and risk appetite respond negatively to policy surprises, while the VIX, the dollar, the BBB spread, the CDX index, and MBS yield and spread respond positively. These results are qualitatively consistent with earlier findings in the literature (Gürkaynak et al., 2005; Bernanke and Kuttner, 2005; Swanson, 2021; Bauer et al., 2023). The estimates indicate that policy surprises lower cash flow expectations, raise risk premia, or both, in addition to their effect on risk-free discount rates. The response is especially strong for the stock market, dollar, risk appetite index, and MBS yield. The magnitudes are economically meaningful; for example, the S&P 500 declines by one percent for a surprise corresponding to a 10 basis point increase in the one-year Treasury yield.

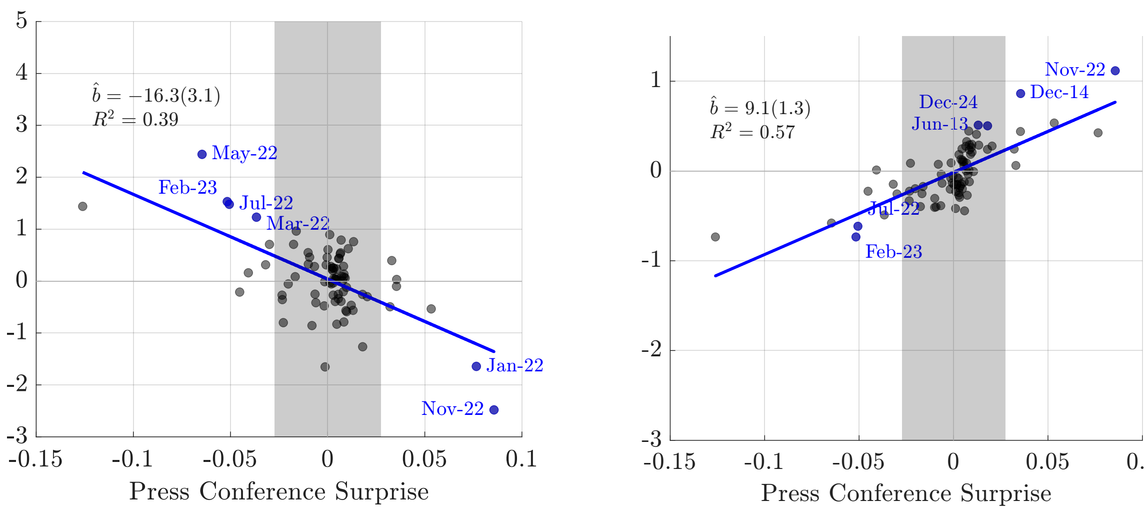
The responses to press conferences, in the second panel of Table 3, tend to be even stronger than for FOMC statements, with larger coefficients and R^2 in most cases.[22] The results for FOMC statements and press conferences in panels (A) and (B) are, however, not directly comparable because press conferences only started in 2011 and thus the sample periods are different. Online Appendix D provides further comparisons of the effects of these different types of policy news using multivariate regressions for the sample of FOMC meetings with press conferences, including the target and path factors of Gürkaynak et al. (2005). These additional results show that when the Committee communicates policy news via both a statement and press conference, the latter is generally the most important driver of risk asset movements.

The responses to the monetary event surprises, covering both FOMC statements and press conferences, are also generally stronger than for the FOMC statement surprises alone. The coefficients for these event-study regressions, shown in panel (C) of Table 3, are statistically significant at conventional levels, with the only exception being the response of the MBS OAS. Similarly to the results for the Treasury market, this evidence confirms that the press

[22]The strong stock market response to press conference surprises is consistent with the finding in Swanson and Jayawickrema (2024) that forward guidance surprises around press conferences have a strong negative impact on the S&P 500.

Figure 3: Intraday risk assets and press conference surprises

(A) S&P 500 *(B) Dollar*



Intraday returns in risk assets (vertical axis, in percent) and press conference surprises (horizontal axis, in percentage points). Panel (A): S&P 500 returns. Panel (B): returns on dollar index (DXY). Sample: FOMC press conferences from 2011 to 2024 (80 observations). Six most influential observations—with the most negative (positive) *DFBETA* for the stock market (dollar) regressions—highlighted and labeled. Text boxes report regression results, including White standard errors. Gray-shaded areas correspond to one standard deviation, above and below zero, for the press conference surprise.

conferences contain information that is relevant for the prices of risk assets and goes beyond what was disclosed in the FOMC statements alone.

With the exception of the dollar, we find no systematic reaction of risk assets to minutes surprises. Taken together with the earlier results for the Treasury market, and the fact that minutes surprises generally have low volatility, the evidence overall suggests that the minutes contain only limited market-relevant new information.

We now turn our focus to the response of the S&P 500 and the dollar. For these two risk assets—arguably the two most important ones in global financial markets—the USMPD contains high-frequency returns around FOMC events. These returns allow more precise estimates of the effects of FOMC communication. Figure 3 plots the intraday S&P 500 and dollar returns against the press conference surprises. Each panel also includes the estimated regression lines, highlights the six most influential observations, and shows shaded areas covering one standard deviation for the policy surprise. The correlations are very tight, and the signs are consistent with standard channels of monetary transmission: press conference surprises have strong negative effects on the stock market and strong positive effects on the U.S. dollar.[23] The scatter plots in Figure 3 and Online Appendix Figure D.1 show that the estimated comovements are not driven by a small number of outliers. The most important press conference surprises all took place during the 2022–2024 period of monetary tightening. As noted in Section 3.2, during this period, press conferences conveyed information about the policy rate path that was both substantial and largely independent from FOMC statement surprises. Figure 3 confirms the large effects of press conference surprises on risk assets during this episode, consistent with the Chair providing Odyssean forward guidance (Campbell et al., 2012).

Figure 3 also shows that some press conference surprises caused responses of the stock market and the dollar that fall in the "wrong" quadrants. Jarociński and Karadi (2020) and Gürkaynak et al. (2021), who first documented the prevalence of these observations for FOMC statement surprises, interpreted them as instances in which information effects dominated relative to conventional channels. For press conference surprises we find only a few such observations, and they all tend to be associated with relatively small policy surprises, that is, inside the gray-shaded areas corresponding to one standard deviation above and below zero. This more limited occurrence of "wrong-signed" responses could be viewed as surprising, if one takes the view that information effects are especially likely to arise around press conferences.[24]

[23] Online Appendix D reports the intraday regression results for all four event types (Table D.3), which show substantially higher explanatory power than for daily returns, as well as scatter plots for statement and monetary event surprises (Figure D.1).

[24] In the presence of information effects the size of any single asset-price movement is not informative about which underlying structural shock is causing the market reaction. While event-study regressions at daily or higher frequency tend to recover average effects consistent with standard macroeconomic theory, caution should be used when extrapolating from this evidence. The same monetary policy surprises can still display puzzling responses of macroeconomic aggregates and survey forecasts when used to proxy for monetary policy shocks in systems estimated at lower frequencies (Miranda-Agrippino and Ricco, 2021; Acosta, 2026).

5.2 Dividend Derivatives

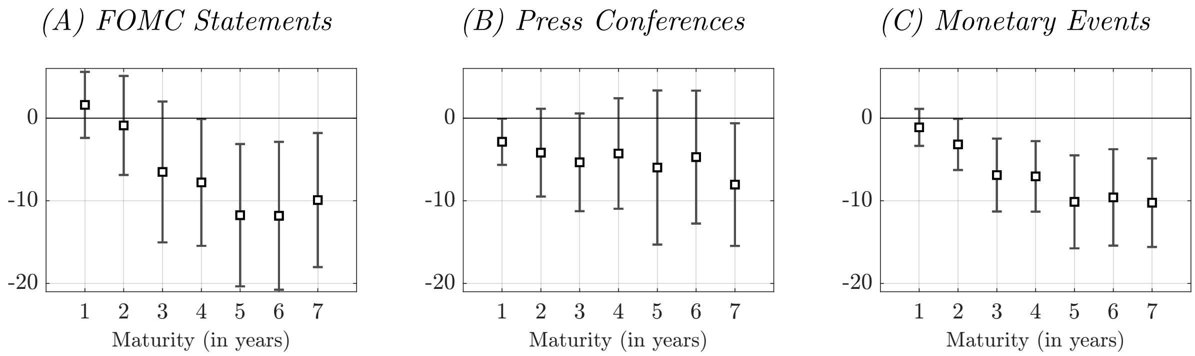
Derivatives tied to future dividends can provide evidence on the reaction of dividend expectations to changes in monetary policy and yield new insights about the channels of monetary transmission. Dividend derivatives—dividend strips, swaps, and futures—are therefore particularly promising to analyze in event studies of FOMC communication.

We estimate the effects of monetary policy on dividend expectations using data on dividend swaps and futures. These instruments have important advantages over dividend strips, which are based on option prices. First, they are tied directly to future dividends, while dividend strip prices are backed out from option prices and put-call parity, raising serious concerns about measurement error. Boguth et al. (2023) show that the synthetic prices of dividend strips magnify the measurement errors in the underlying option prices. Second, they have annual expirations out to several years, which allows us to obtain meaningful term structure evidence, while reliable option prices are available only out to a horizon of a few months. Third, the prices of dividend swaps and futures are (essentially) forward prices of dividends, while the derived spot prices of dividend strips need to be further translated into market-based dividend expectations by accounting for changes in risk-free interest rates. The literature on the term structure of equity risk premia has also generally moved towards the use of dividend swaps and futures data (van Binsbergen and Koijen, 2017).

Dividend swaps and dividend futures are very similar instruments in how their payoffs are tied directly to future dividends, and indeed the terms are sometimes used interchangeably in the asset pricing literature. Dividend swaps are over-the-counter derivatives between financial intermediaries and have been used in the U.S. since the early 2000s. Futures have the advantage of being standardized exchange-traded contracts, but for the U.S. they have only been trading since late 2015, resulting in shorter samples. We focus on dividend swaps due to the longer available sample period.

Our dividend swap data is from Goldman Sachs and consists of daily prices over a sample period starting in 2004. Dividend swap contracts for the U.S. are tied to dividends on the

Figure 4: Response of dividend swap returns to monetary policy surprises



Event-study regressions of two-day returns in dividend swaps, for constant (interpolated) annual horizons, on different high-frequency monetary policy surprises. Panel (A): responses to statement surprises. Panel (B): responses to press conference surprises. Panel (C): responses to monetary event surprises. The sample includes FOMC announcements from 2010 to 2024 (panels (A) and (C)), or press conferences from 2011 to 2024 (panel (B)). Unscheduled FOMC meetings whose daily event window spans a market closure are excluded. Error bars show 95% confidence intervals based on White standard errors.

S&P 500 over each future calendar year (Manley and Mueller-Glissmann, 2008). This data has been used extensively in related research (e.g., van Binsbergen and Koijen, 2017; Nagel and Xu, 2026). We calculate two-day log returns (in percent) to account for the potentially slower transmission of policy surprises to the price of assets with lower liquidity. We construct constant-maturity data by linearly interpolating these returns.[25] Our baseline sample period starts in 2010 due to low liquidity during earlier years of the dividend swap market and during the financial crisis. Online Appendix D reports additional results for other sample periods and also for dividend futures.

Figure 4 shows the estimated responses of dividend swap returns to FOMC surprises for horizons from one to seven years. The estimated response coefficients are generally negative, the only exceptions being some short-horizon coefficients that are positive but statistically insignificant. For FOMC statements, the negative coefficients are statistically significant for horizons of four years and longer. The response coefficients for press conferences are all negative but generally not statistically significant. Nevertheless, the press conferences

[25]Our method of interpolation is similar to that of van Binsbergen and Koijen (2017). For example, if t is the current trading day, F_{1t} is the return on the current-year contract with last trading day T at the end of December, F_{2t} is the return on the subsequent contract, and w is $T - t$ measured in years, then we construct the return on a one-year dividend contract as $w\, F_{1t} + (1 - w)\, F_{2t}$.

materially strengthen the negative responses, with the result that the coefficients for monetary event surprises are strongly statistically significant for horizons of three years and longer. Online Appendix D reports the corresponding regression results, the estimates for the dividend swap responses to minutes surprises, and further robustness checks showing that while the estimated response varies somewhat across sample periods, the overall patterns are robust.[26]

Our estimates suggest a negative response of dividend expectations to monetary policy surprises. The effects are strongest at longer horizons, similar to the term structure patterns for breakeven inflation in Section 4. As discussed there, these patterns are more likely to reflect the various types of policy news captured by our surprises rather than point to long transmission lags of monetary policy.[27]

The patterns in our estimates may appear to contrast with the results in other recent studies, but there are important differences in the horizons and data. Gilchrist et al. (2024) and Golez and Matthies (2025) both estimated significant positive effects of FOMC statement surprises on short-term dividend strip prices, which they interpreted as evidence of information effects dominating conventional channels. The positive effects on dividend strips were documented for very short horizons up to 180 days, while we estimate negative effects for horizons of several years. We also estimate positive (though insignificant) response coefficients at the shortest horizon, so the patterns in dividend strip and swap prices are not necessarily inconsistent. The measurement-error concerns for dividend strips discussed above make a direct comparison of the two sets of estimates difficult, since noise in the strip prices could obscure their true response to policy surprises (Boguth et al., 2023). In any event, information effects at short horizons could coexist with the conventional effects that dominate at the longer horizons we study.

[26]Results for dividend futures are qualitatively similar. For the 2005–2024 sample of dividend swaps, coefficients remain negative but are much smaller and statistically insignificant, likely due to low liquidity during the early years of this market. Scatter plots visualize the robustness of our estimates, and excluding one large outlier has no material impact on the results.

[27]The interpretation through the lens of transmission lags is somewhat complicated here by the fact that dividend derivative prices capture *levels* of future dividends, while questions about the lags of monetary policy are generally focused on the effects on *growth rates*. In addition, these derivative prices depend on *nominal* future dividends, which compound any expected effects on inflation.

Our results are consistent with the standard channels of monetary transmission that predict a negative impact on corporate profits and dividends. Bilbiie and Känzig (2023) show that both in the data and in the standard New Keynesian model, corporate profits respond negatively to a contractionary monetary policy shock. The term structure patterns also conform with the stylized fact from corporate finance that dividends are sticky (Lintner, 1956; Leary and Michaely, 2011).

An implication of our evidence is that changes in risk-adjusted dividend expectations—including actual expectations and cash flow risk premia—contribute to the negative stock market response to monetary policy surprises. In related work, Nagel and Xu (2026) use dividend futures to calculate the stock market response implied by changes in the yield curve alone and find that a large share of the actual stock market response to FOMC statement surprises can be attributed to this channel. While we emphasize an alternative channel, our results do not contradict their statistical evidence on the Fed's impact on the stock market.[28] The evidence suggests that both changes in risk-free rates and in market-based dividend expectations play a meaningful role in the transmission of monetary policy to the stock market. We leave it to future work to quantify the relative importance of these channels.

6 Conclusion

This paper introduces the USMPD, a new public database of high-frequency monetary policy surprises around FOMC communication events. Unlike other available sources, the USMPD includes data on a broader set of FOMC communication events—including statements, minutes, and press conferences—and a wider range of risk-free rates and risk asset prices. The USMPD is updated regularly using a consistent and transparent methodology. Our hope is that the availability of this data source of U.S. monetary policy surprises will further

[28] The importance of the residual response, due to changes in risk premia and dividend expectations, varies across specifications estimated by Nagel and Xu (2026), in some cases accounting for roughly 30% of the total stock market response. The confidence intervals around this response are wide, leaving room for a material contribution from the channel we emphasize.

stimulate empirical research in monetary economics.

The paper establishes several new facts about monetary policy news and their financial market effects. Monetary policy surprises have increased in magnitude over recent years, providing renewed relevance for the use of high-frequency event studies and identification methods for empirical analyses of monetary policy. Press conferences convey policy news that is largely independent of the statements, with strong effects on Treasury yields and risk asset prices. Disregarding them therefore misses a substantial part of FOMC policy news, and we recommend the monetary event surprise that spans both events. In contrast to earlier studies, our estimates show that monetary policy actions and communication have significant negative effects on market-based expectations of both future inflation and dividends, reconciling the financial market evidence with the disinflationary effects of monetary policy in standard macro models and time series estimates. Across asset classes, the sign of the financial market effects of FOMC communication is generally consistent with standard monetary transmission channels. At the same time, the term structure patterns of these effects peak at longer horizons than standard monetary transmission would predict, suggesting that monetary policy surprises also convey signals that affect beliefs about the future conduct of monetary policy.

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